Goal: Be able to undo the process of differentiation.

$\int kf(x)\ dx = kF(x) + C$	$\int (f(x) + g(x))\ dx = F(x) + G(x) + C$
$\int f(g(x))g'(x)\ dx = F(g(x)) + C$	$\int (f(x) - g(x))\ dx = F(x) - G(x) + C$
$\int x^n\ dx = \frac{1}{n+1}x^{n+1} + C$ (when $n \neq -1$)	$\int \frac{1}{x}\ dx = \ln\lvert x\rvert + C$
$\int e^x\ dx = e^x + C$	$\int e^{kx}\ dx = \frac{1}{k}e^{kx} + C$
$\int \sin x\ dx = -\cos x + C$	$\int \cos x\ dx = \sin x + C$
$\int \sec^2 x\ dx = \tan x + C$	$\int \sec x \tan x\ dx = \sec x + C$
$\int \csc x \cot x\ dx = -\csc x + C$	$\int \csc^2 x\ dx = -\cot x + C$
$\int \frac{1}{\sqrt{1-x^2}}\ dx = \sin^{-1} x + C$	$\int \frac{1}{1+x^2}\ dx = \tan^{-1} x + C$

Table 1: Table of Antiderivatives

Antiderivatives

Definition. A function F is an _____ of f on an interval I if $F'(x) = f(x)$ for all x an I. (If I is not specified, then it is just the domain of f).

Example 1. Consider the function $f(x) = x$:

1. Is $F(x) = \frac{1}{2}x^2$ an antiderivative of x (on $I = (-\infty, \infty)$)?

2. Is $F(x) = x^2$ an antidervative of x?

3. Is $F(x) = \frac{1}{2}x^2 + 1$ an antiderivative of x?

Conclusion:

Theorem. If $F' = 0$ on an interval, then F is _____ on the interval.

WHY?

Corollary. Corollary If F and G are antiderivatives of a function f, then

$$F(x) =$$

for some _____ C.

WHY?

Indefinite Integration

Definition. The family of antiderivatives of a function f is called the _____ of f, denoted

$$\int f(x)\,dx \overset{\text{BY THE COROLLARY}}{=\!=\!=}$$

Example 2. Find all possible antiderivatives of $f(x) = \sin x$.

Example 3. Compute $\int \frac{1}{x}\, dx$ on $(0, \infty)$

Theorem. In general, on $(-\infty, 0) \cup (0, \infty) = \left\{\text{domain of } \frac{1}{x}\right\}$

$$\int \frac{1}{x}\, dx =$$

Example 4. Compute $\int x^2\, dx$

Theorem (Reverse Power Rule)**.** When $n \neq -1$

$$\int x^n dx =$$

Example 5.

$\int \sqrt{x}\, dx =$

Theorem. Integrals are linear in the sense that:

(i) $\displaystyle\int (f(x) \pm g(x))\ dx = \int f(x)\ dx \pm \int g(x)\ dx$

(ii) If k is a constant, $\displaystyle\int kf(x)\ dx = k\int f(x)\ dx$

Example 6. Compute $\displaystyle\int \frac{2x^5 - \sqrt{x}}{x}\ dx$

Remark 1 (Notation): Add parentheses when the integrand has more than one term:

$$\int x^2 + x^3\ dx \longleftarrow \text{Not Quite Right}$$

$$\int (x^2 + x^3)\ dx \longleftarrow \text{Correct}$$

Remark 2: When dealing with multiple integrals, it is common practice to have only one constant of integration and to tack it on at the very end. We will demonstrate why in the next example.

Example 7. Compute $\int \sec x(\sec x + \cos x)\ dx$

Example 8. Compute $\int \dfrac{14 + 4xe^x}{x}\ dx$

A Glimpse into MA 16020

We want techniques to "undo" the use of the Product, Quotient, and Chain Rule. **You won't need this for 16010**, but here is an example of such a thing:

Theorem (Reverse Chain Rule/"U-substitution")**.** If F is an antiderivative for f. Then,

$$\int f(g(x)) \cdot g'(x)dx = F(g(x)) + C$$

Example 9. Compute $\int \cos(2x)\ dx$